

Jaume **Ivars Grimalt**

Computer Vision Engineer

Zürich, Switzerland, open to relocation

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Profile

I train detection architectures from scratch on our own data, at scale on HPC, and ship them to production. I cover the whole path: the model, the training system, and the product it runs in. At Mycrospace that took high-density colony counting from a Master's thesis to a product tested in **5 labs**.

Technical Skills

Computer Vision	Object detection, instance segmentation, density estimation, classification
Training	Python, PyTorch (DDP, AMP, torch.compile), Hugging Face, timm, multi-GPU training on HPC (SLURM, checkpoint-resumed jobs)
Production	ONNX, Docker, GCP (Cloud Run, Cloud SQL, GCS), Hetzner + Dokploy, BullMQ/Redis, CI/CD
Product Engineering	NestJS, Prisma, PostgreSQL, MongoDB, Next.js, TypeScript, React Native/Expo, Stripe, PostHog

Experience

Mycrospace — *Zürich, Switzerland*

Co-Founder & CTO — *Jan 2024 -- Sep 2026*

- Computer vision for microbiology. Started in 2024 as my Master's thesis; the company was incorporated in January 2026.
- **Modelling & research.** Developed our own detector with ConvNeXt architecture inspired by YOLO and DQ-DETR, trained from scratch on our own data.
- Flexible detection from 10 to 2000 objects per image.
- Colony detection at **62.4 mAP50, 42.5 mAP50-95**. Validated against expert microbiologists: 75% recall and 90% precision.
- **Training & HPC.** More than **12,500 GPU-hours** on the Leonardo supercomputer under two EuroHPC AI Factory compute allocations in the computational biology domain. Multi-GPU training on **4x A100** with DDP, AMP and torch.compile under SLURM, with chained jobs that resume from checkpoint when wall clock runs out.
- Own training framework, not third-party scripts. Cross-validation and ensembles with weighted box fusion.
- **1,500** images annotated out of **5,000** across several species under protocols written with microbiologists. I developed our own weakly-supervised labelling tools: click → crop → 4-channel CNN with a Gaussian heatmap → bbox regression. Annotation versioning so the model improves as labels improve thanks to scoring images with k-folding.
- **MLOps & infrastructure.** Containerised inference with queue workers on BullMQ and Redis; image upload via signed URLs and bucket triggers; real-time events over SSE/WebSocket.
- It runs on GCP (Cloud Run, Cloud SQL, GCS and Cloud Scheduler), product monitoring with PostHog, and internal LLMs for automations.
- **Product engineering.** NestJS/Prisma backend over PostgreSQL; MongoDB for annotations. Own schema design. Next.js/TypeScript web client and a React Native/Expo app for the bench.
- Stripe billing: subscriptions, webhook lifecycle, B2B invoice flows, credit renewal.

MIALAB -- Medical Imaging Analysis Laboratory, UPV — *Valencia, Spain*

Research Engineer — *May 2025 -- Jan 2026*

- Volumetric brain MRI segmentation with nnU-Net, comparing multi-scale feature-extraction strategies.
- VolBrain has processed **700k+** MRI volumes, so inference cost scales. Profiled the pipeline from NIfTI ingest to voxel-level prediction to find where the time went.

Tapstar — *Valencia, Spain*

Full-Stack Engineer (freelance) — *Jan 2025 -- May 2025*

- QR generation, back office, and frontend for QR cards and stands sold to high-street businesses; scan analytics and per-employee ratings.

Vesta-Z — *Valencia, Spain*

Data Engineer — *Jul 2023 -- Feb 2025*

- Full-stack feature work on their cloud product; custom schema for cross-matching items across providers; LLM customer-support bot.

Neurocatching — *Valencia, Spain*

Data Scientist Intern — *Feb 2023 -- Aug 2023*

- Eye-gaze model training for neuromarketing studies; data analysis for customer behaviour and sales.

Sciling — *Valencia, Spain*

Backend Developer Intern — *Jun 2022 -- Sep 2022*

- Conversational systems on GPT-3-era LLMs, applied NLP, and early diffusion models.

Projects

ConvNeXt-V1 trained from scratch

Own training framework on EuroHPC (Leonardo Booster)

- Trained on **4x A100** on Leonardo Booster with a training framework.
- ConvNeXt-Tiny, **28M** parameters, **81% top-1** on ImageNet-1k over **300 epochs** with timm augmentation (82.1% reported in the original paper). A close replication, which is the point: it gives me a baseline I own. github.com/jaume2000/convnext_v1
- In progress on that baseline: redundant-parameter search, and reparametrisation of residual blocks interpreted as steps of an Euler solver.

Publications

- *Auto-N2N: Self-supervised MRI Denoising using Synthetic Pair Generation and Mixture of Experts.* Sergio Morell-Ortega, Alexey Mengliev, Angela Gonzalez-Cebrian, **Jaume Ivars Grimalt**, Pierrick Coupe, Jose V. Manjon. Preprint, 2026. OpenReview

- *Detecting Cytopathic Effect in Cell Culture with Computer Vision.* Walter Ragazzo, **Jaume Ivars Grimalt**. Under review.

- *One Block, 10,000 Steps: ConvNeXt Stages as Nearly Autonomous Vector Fields.* **Jaume Ivars Grimalt**. In preparation.

Education & Awards

M.Sc. in Artificial Intelligence, Pattern Recognition & Digital Imaging — *Valencia, Spain*

Universitat Politècnica de València · GPA 8.7/10 · Thesis awarded Honours, and it became Mycrospace — *2023 -- 2024*

Technology Management (add-on programme) — *Valencia, Spain*

Center for Digital Technology and Management (CDTM) · TUM/LMU Munich, Valencia campus · Second Valencia cohort, 16 students selected — *2024*

B.Sc. in Computer Engineering — *Valencia, Spain*

Universitat Politècnica de València · GPA 8.3/10 · Honour mentions in Machine Learning and Statistics — *2019 -- 2023*

1st Prize, IdeasUPV Startup Competition — *Valencia, Spain*

Mycrospace, 2500 EUR grant — *2024*

Languages

Spanish (native), Catalan (native), English (C1)